Design Verification Plan and Report (virtual test) — VBF-T1R1-DVPR-01

Case: t1_r1 | Date: 2026-08-25 | Prepared: ____________ | Reviewed: ____________ | Approved: ____________

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1C discharge capacity	run-pyamm --protocol 1C_discharge, SPMe, 298.15 K	5.6505 Ah	PASS vs parameter nominal 5.0 Ah (no registered capacity criterion)	cell/r3_I_1c.json:capacity_ah
4C fast-charge temperature rise	run-pyamm --protocol 4C_charge_45C, DFN, lumped thermal, T_amb 318.15 K	T_max 325.655 K = 52.51 °C, rise +7.50 K	PASS vs criterion <= 333.15 K (margin 7.50 K)	cell/r3_I_4c_dfn.json:T_max_K
4C fast-charge plating	same protocol + plating module; negative electrode potential vs Li/Li+	min anode potential +0.0249 V (>= 0 -> plated = false)	PASS vs criterion plated = false (margin 24.9 mV)	cell/r3_I_4c_dfn.json:anode_potential_v (DFN authoritative; SPMe corroborates +0.0135 V in cell/r3_I_4c.json)
Voltage window	parameter set limits	2.5 – 4.2 V	PASS (design envelope per Chen2020)	Chen2020 cut-off parameters
Overcharge without thermal runaway	run-pyamm --protocol overcharge (0.5C to 4.2 V + 0.5 V = 4.7 V), then run-tr with mass-kg from calc-energy	terminal voltage 4.7000 V; overcharge peak 299.552 K; TR triggered = false	PASS vs criterion triggered = false	cell/r3_I_oc.json (voltage tail, T_max_K); cell/r3_I_tr.json:triggered
Nail penetration	—	N/A (requires physical experiment)	N/A (beyond pure simulation boundary, requires physical experiment)	—
Crush / drop	—	N/A (requires physical experiment)	N/A (beyond pure simulation boundary, requires physical experiment)	—
Cycle life	—	N/A (no aging model in this case; aging protocol not part of registered criteria)	N/A (beyond pure simulation boundary)	—
Rate-pulse internal resistance	—	N/A (requires physical experiment)	N/A (beyond pure simulation boundary; DC resistance proxy available: dcr_ohm 0.155 m?, cell/r3_I_energy.json)	—

Conclusion

- PASS summary: energy density 553.977 Wh/kg (criterion >= 392.61, round-4 evaluate PASS); 4C fast charge without plating (DFN authoritative, anode min +0.0249 V); T_max 325.655 K <= 333.15 K; overcharge 4.7000 V without thermal runaway (triggered = false). All four registered stage2/stage3 criteria satisfied by candidate r3-I.
- Per-round funnel record: round 0 baseline fail (plating -0.4385 V); round 1 architecture all fail (plating); round 2 formulation fallback, Tmax solved, plating -0.0149 V; round 3 all four candidates pass, finalist r3-I (margins: ED +161 Wh/kg, Tmax 7.50 K, anode potential 24.9 mV).
- Uncovered items (paper limitations): molecular-layer criteria (max_energy_ev / max_homo_ev) unchecked — no molecular candidates proposed, real_compute = false; nail penetration / crush / drop / cycle life N/A (requires physical experiment).